TUSHAR MOPURI

Fourth-Year PhD Student ◇ Website
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EDUCATION

University of Pennsylvania *August 2023 - Present*
PhD in Computer and Information Science
Advisor: Pratyush Mishra

Indian Institute of Science *August 2018 - April 2022*
BS (Research)
Major: Mathematics

PREPRINTS

Authors are listed in alphabetical order (by last name), unless otherwise noted.

Vector Commit-and-Reduce Schemes
Anubhav Baweja, Pratyush Mishra, Tushar Mopuri, Bharath Namboothiry
(ePrint)

CONFERENCE PUBLICATIONS

Field-Agnostic SNARKs with Small Proofs via Encode-Repeat-Accumulate (ERA) Codes
Anubhav Baweja, Giacomo Fenzi, Pratyush Mishra, Tushar Mopuri
ASIACRYPT 2026
Also presented at ZKProof 8
(ePrint)

FICS and FACS: Fast IOPPs and Accumulation via Code-Switching
Anubhav Baweja, Pratyush Mishra, Tushar Mopuri, Matan Shtepel
CRYPTO 2026
Also presented at ZKProof 8 (merged)
(ePrint)

Query-Optimal IOPPs for Linear-Time Encodable Codes
Anubhav Baweja, Pratyush Mishra, Tushar Mopuri, Matan Shtepel
EUROCRYPT 2026
Also presented at ZKProof 8 (merged)
(ePrint)

Scribe: Low-memory SNARKs via Read-Write Streaming
Anubhav Baweja, Pratyush Mishra, Tushar Mopuri, Karan Newatia, Steve Wang
USENIX Security 2026
Runner-Up for Distinguished Paper Award
Also presented at ZKProof 8 and zkSummit 11
(ePrint)

Time-Space Trade-Offs for Sumcheck
Anubhav Baweja, Alessandro Chiesa, Elisabetta Fedele, Giacomo Fenzi, Pratyush Mishra, Tushar Mopuri, Andrew Zitek-Estrada
TCC 2025
Also presented at ZKProof 8 and zkSummit 14
(ePrint)

DewTwo: a transparent PCS with quasi-linear prover, logarithmic verifier and 4.5KB proofs from falsifiable assumptions
Benedikt Bünz, Tushar Mopuri, Alireza Shirzad, Sriram Sridhar
CRYPTO 2025
(ePrint)

Hekaton: Horizontally-Scalable zkSNARKs via Proof Aggregation
Michael Rosenberg, Tushar Mopuri, Hossein Hafezi, Ian Miers, Pratyush Mishra (not alphabetical)
ACM CCS 2024
(ePrint)

Dew: Transparent Constant-sized zkSNARKs
Arasu Arun, Chaya Ganesh, Satya Lokam, Tushar Mopuri, Sriram Sridhar
PKC 2023
(ePrint)

RESEARCH VISITS

Simons Institute, Berkeley - Attended cryptography workshops *Summer 2025*
New York University - Worked with Benedikt Bünz *Summer 2024*
Aarhus University - Worked with Claudio Orlandi, Peter Scholl and Lawrence Roy *Fall 2022*
Microsoft Research (India) - Interned with Satya Lokam *Spring 2022*

FELLOWSHIPS

Jonathan M. Smith Fellowship *August 2023 - July 2024*
Awarded by the University of Pennsylvania.

Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship *August 2018 - April 2022*
Awarded by the Department of Science and Technology (DST), Govt. of India.

TEACHING AND REVIEWING

I was a TA for CIS 5560 (Cryptography) at Penn in Spring 2024 and Spring 2025. Additionally, I was an external reviewer for Eurocrypt 2025, Crypto 2025, TCC 2026, Asiacrypt 2026 and USENIX Security 2027.